Networking Engineering — Phase 0 Study Guide

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Phase: 0 — Environment Readiness and Networking Foundations

Status: Completed

Assessment: 10/10 (100%)

Portfolio: GitHub Repository

1. Phase 0 Objectives

By completing this phase, you have learned to:

- Inspect a Windows computer's network configuration.
- Understand IP addresses, subnet masks and default gateways.
- Explain DHCP and DNS.
- Build a basic LAN in Cisco Packet Tracer.
- Understand Ethernet switching and MAC address learning.
- Inspect Ethernet frames.
- Explain ARP and address resolution.
- Calculate IPv4 subnets.
- Test connectivity and interpret failures.
- Document practical networking projects in GitHub.

2. Your Learning Environment

Host computer

Component	Specification
Operating system	Windows 10 Pro 22H2
Processor	Intel Core i5-4300M
RAM	8 GB
Storage	232 GB
Virtualization	Enabled
VirtualBox	7.2.24

Networking software

- **Cisco Packet Tracer 9.0.1** — installed and working.
- **Wireshark** — not yet installed.
- **VirtualBox** — installed.

3. Basic Network Configuration

You inspected your Windows network configuration using:

```
ipconfig
```

To display additional adapter information:

```
ipconfig /all
```

Important networking terms

Term	Meaning
IPv4 address	Identifies a device on an IP network
Subnet mask	Separates network and host portions
Default gateway	Forwards traffic to other networks
DNS	Resolves domain names into IP addresses
DHCP	Automatically supplies network configuration
MAC address	Identifies a network interface at Layer 2

DHCP

DHCP automatically assigns network configuration. Its four main stages are called **DORA**:

1. Discover
2. Offer
3. Request
4. Acknowledge

DNS

DNS translates names such as `example.com` into IP addresses. You tested DNS using:

```
nslookup example.com
```

Connectivity testing

You tested connectivity using:

```
ping 192.168.10.1
```

(The address above is illustrative, not your actual home-network address.)

You successfully tested your local gateway and an external IP address during Phase 0.

> **Remember:** A successful ping proves that an ICMP response was received. It does not prove that every network service is working.

4. Building Your First LAN

You created a Packet Tracer topology containing:

- Three PCs.
- One Cisco 2960 switch.
- Copper straight-through Ethernet cables.

Original LAN addressing

Device	IPv4 address	Subnet mask
PC0	`192.168.1.10`	`255.255.255.0`
PC1	`192.168.1.11`	`255.255.255.0`
PC2	`192.168.1.12`	`255.255.255.0`

All three PCs communicated successfully within the same subnet.

Important lesson

Devices in the same IP subnet can communicate through a Layer 2 switch without requiring a default gateway for that local communication.

5. Ethernet Switching and MAC Learning

An Ethernet switch learns MAC addresses by examining the source MAC address of incoming frames. It stores the learned address and the port where the frame arrived.

You inspected the switch using:

```
show mac-address-table
```

You observed dynamically learned MAC addresses associated with switch ports.

How a switch forwards frames

1. A frame arrives on a switch port.
2. The switch learns the source MAC address.
3. It examines the destination MAC address.
4. If the destination is known, it forwards the frame through the associated port.
5. If the destination is unknown, it floods the frame through eligible ports in the same VLAN, excluding the incoming port.

Important distinction

A Layer 2 switch forwards Ethernet frames using MAC addresses. A router forwards packets between IP networks using routing information.

6. Ethernet Frames

An Ethernet frame contains:

1. Destination MAC address
2. Source MAC address
3. EtherType
4. Payload
5. Frame Check Sequence (FCS)

You used Packet Tracer Simulation Mode to inspect traffic between PCs. You observed the frame's source and destination MAC addresses and followed the packet through the switch.

Remember

- **Source MAC:** where the Ethernet frame came from.
- **Destination MAC:** where the frame is intended to go.
- **IP address:** identifies the logical network destination.
- **MAC address:** identifies the local Ethernet interface destination.

7. ARP — Address Resolution Protocol

ARP helps a device discover the MAC address associated with a known IPv4 address on its local network.

ARP process

1. PC0 knows PC1's IPv4 address.
2. PC0 does not know PC1's MAC address.
3. PC0 broadcasts an ARP Request.
4. PC1 recognizes its IP address and sends an ARP Reply.
5. PC0 stores the mapping in its ARP cache.

You inspected ARP information using:

```
arp -a
```

You observed dynamic IP-to-MAC mappings.

Important distinctions

- **ARP Request:** normally broadcast.

- **ARP Reply:** normally unicast.
- **ARP cache:** stores recently learned mappings.
- **Ethernet broadcast MAC:** `FFFF.FFFF.FFFF` in Packet Tracer's display format.

You also followed an ARP Request from PC0 through the switch to PC2 and observed PC2's reply.

8. IPv4 Addressing

IPv4 addresses contain 32 bits, divided into four 8-bit octets (e.g., `192.168.1.10`). Each octet can range from 0 to 255.

Binary place values

Bit position	Decimal value
1st	128
2nd	64
3rd	32
4th	16
5th	8
6th	4
7th	2
8th	1

Examples

Binary	Decimal
`10000000`	128
`11000000`	192
`11100000`	224
`11110000`	240
`11111000`	248
`11111100`	252
`11111110`	254
`11111111`	255

CIDR notation

CIDR notation tells us how many bits belong to the network portion.

Prefix	Subnet mask	Total addresses	Usable hosts*
`/24`	`255.255.255.0`	256	254
`/25`	`255.255.255.128`	128	126
`/26`	`255.255.255.192`	64	62
`/27`	`255.255.255.224`	32	30

*For conventional IPv4 subnets, excluding the network and broadcast addresses.

Network and broadcast addresses

For `192.168.1.0/24`:

- **Network address:** `192.168.1.0`
- **Usable host range:** `192.168.1.1` – `192.168.1.254`
- **Broadcast address:** `192.168.1.255`

Subnet block size

A useful method for common subnet calculations is:

```
Block size = 256 − interesting subnet-mask octet
```

For `/26`:

```
256 − 192 = 64
```

The subnet boundaries are `.0`, `.64`, `.128`, and `.192`.

9. Your Subnetting Practical

You created a separate Packet Tracer file so your original LAN remained untouched.

Addressing plan

Device	IPv4 address	Subnet
PC0	`192.168.1.10/26`	`192.168.1.0/26`
PC1	`192.168.1.70/26`	`192.168.1.64/26`
PC2	`192.168.1.71/26`	`192.168.1.64/26`

Subnet mask for all devices: `255.255.255.192`

Results

- PC1 successfully pinged PC2.
- PC0 could not ping PC1 because they belonged to different IP subnets and no router was configured.

Important lesson

A Layer 2 switch does not automatically route traffic between different IP subnets. A router or Layer 3 switch is needed to route between those networks.

All three devices remained connected to the same physical switch and VLAN 1. Different IP subnets do not automatically mean separate VLANs.

10. Troubleshooting Commands

Command	Purpose
`ipconfig`	Display Windows IP configuration
`ipconfig /all`	Display detailed adapter configuration
`ping`	Test ICMP connectivity
`nslookup`	Test DNS name resolution
`arp -a`	Display ARP cache
`show mac-address-table`	Inspect switch MAC learning

Packet Tracer supports a limited command set. Some commands available on real Cisco equipment may not work in its simulated PC or switch environment.

11. Phase 0 Knowledge Assessment

You completed the final assessment with **10 correct answers out of 10**.

Topics assessed:

- Network address calculation.
- Subnet identification.
- Switch forwarding.
- ARP requests.
- Default gateways.
- DNS troubleshooting.
- Physical connectivity troubleshooting.
- Routing between different subnets.
- MAC address learning.

12. Portfolio Files

Your Phase 0 GitHub repository contains:

- `README.md`
- `.gitignore`
- `phase-0-network-checks.md`
- `phase-0-first-lan-report.md`
- `phase-0-switching-and-mac-learning.md`

- `phase-0-ethernet-frames-and-forwarding.md`
- `phase-0-arp-and-address-resolution.md`
- `phase-0-ipv4-addressing-and-subnet-masks.md`
- `phase-0-first-lan.pkt`
- `phase-0-subnetting-practical.pkt`

Repository: https://github.com/Constantinople-Atlas/Networking-lab *(Currently private)*

13. What Comes Next?

Phase 1 — Network Models and Encapsulation

You will study:

1. Introduction to Network Models
2. The OSI Model
3. The TCP/IP Model
4. OSI vs TCP/IP
5. Data Encapsulation and Decapsulation
6. Protocol Data Units (PDUs)
7. Networking Devices and Their Layers
8. Practical Packet Tracer Lab and Assessment

Phase 1 completion goals

By the end, you should be able to:

- Explain all seven OSI layers in order.
- Explain the TCP/IP model.
- Compare OSI and TCP/IP.
- Describe encapsulation and decapsulation.
- Distinguish segments, packets, frames and bits.
- Identify the layers at which common networking devices operate.
- Interpret basic Packet Tracer network events.

Phase 0 completed successfully. Continue learning one practical step at a time, verify every result, and document each milestone in your Networking GitHub portfolio.